



LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



OpenCTI curation: Normalize IP-like labels

Threat Intelligence Report

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Table of Contents

1. Executive Summary
2. Intelligence Requirement
3. Source Review and Confidence
4. Key Findings
5. Threat Context
6. Detection and Hunting
7. Recommendations
8. References

1. Executive Summary

Lost Boys Cyber reviewed OpenCTI curation candidate `opencti-curation-8a69c93a75eb2cad`, which flags 30 sampled taxonomy labels that encode IPv4 addresses with dashes instead of dots. The review confirms the candidate is valid: each sampled dashed-IP label is attached to one or more OpenCTI URL observables and matching indicators that already contain the dotted IPv4 value in the canonical STIX `url:value` pattern. These strings are operational indicators or observables, not durable taxonomy terms.

The recommended action is controlled normalization, not blind deletion. Remove the dashed-IP labels from objects only after preserving the underlying URL indicators and observables, and create or link standalone `IPv4-Addr` observables only where the IP address itself needs first-class infrastructure modeling beyond the URL object. Because the source candidate is marked `safe\_mode: True` and `manual-review-only`, no destructive change should be applied until the specific label-to-object relationships are exported, reviewed, and reconciled.

Field	Value
Report family	Threat Intelligence Report / OpenCTI Source-of-Truth Curation
Candidate ID	`opencti-curation-8a69c93a75eb2cad`
Category	`label_normalization`
Severity	Medium
Proposed action	Review
Safe mode	True
Source artifact	`/home/lost0x01/services/opencti-deploy/audit/latest-opencti-hygiene-audit.json`
Cleanup candidate artifact	`/home/lost0x01/services/opencti-deploy/audit/latest-opencti-cleanup-candidates.json`
Audit generated	2026-06-16T03:21:38.621399+00:00
Live validation evidence	`/home/lost0x01/.hermes/kanban/workspaces/t_2fe470e3/opencti_ip_label_review_evidence.json`

2. Intelligence Requirement

This review answers whether dashed-IP labels in OpenCTI should remain as taxonomy labels, be converted into STIX observables, or be removed after equivalent observables and relationships are confirmed.

Requirement	Assessment
Determine whether the 30 sampled labels are taxonomy or entities	They are entity-like strings. Each value normalizes cleanly from `octet-octet-octet-octet` to a dotted IPv4 address.
Confirm whether equivalent OpenCTI intelligence objects exist	Confirmed for all 30 sampled values: each dotted IP search returned at least one URL observable and at least one indicator.
Determine safe cleanup path	Remove dashed-IP labels from indicators/observables after relationship export and review; create first-class IPv4 observables only where operationally useful.
Identify defensive value	Use the normalized IP and URL values for detection/hunting; do

	not rely on dashed-IP label strings as detection keys.
Preserve auditability	Keep candidate and evidence JSON artifacts with the task record; do not mutate under `safe_mode` without an apply plan.

Assumption: this report is limited to the 30 sampled IP-like labels in the candidate. It does not assert that every IP-like label in the full OpenCTI corpus has equivalent observables unless covered by a future full-corpus export.

3. Source Review and Confidence

The source-of-truth hygiene audit was non-destructive and reported a healthy OpenCTI API response at the time of generation. The audit sampled 500 labels and found 30 dashed-IP labels plus 12 ASN-like labels; CVE-like labels were not present in that sample. A live GraphQL validation pass was then run against the local OpenCTI API for the 30 candidate values, searching each dotted IP value across `stixCyberObservables` and `indicators`.

Source	Role	Confidence	Notes
`latest-opencti-hygiene-audit.json`	Primary audit source	High	Generated by local hygiene automation; OpenCTI health returned `success`.
`latest-opencti-cleanup-candidates.json`	Candidate queue source	High	Marks this item `manual-review-only` and `safe_mode: true`.
Live OpenCTI GraphQL validation	Corroboration	High	Confirmed observable and indicator coverage for all 30 sampled IP values.
External enrichment	Not used	Not applicable	This is a private source-of-truth hygiene review, not an attribution report.

OpenCTI corpus context from the same audit:

Object class	Count
Reports	768
Indicators	45,767
STIX Cyber Observables	44,204
Labels	4,356
Intrusion Sets	442
Threat Actors	0
Malware objects	4,433
Vulnerabilities	1,735
External References	39,895

Confidence is high that the candidate is a real taxonomy hygiene issue. Confidence is medium that direct deletion is safe without further review, because the live search confirms equivalent URL-level objects but does not prove that every consuming report, grouping, or relationship has been reviewed for label dependency.

4. Key Findings

ID	Finding	Severity	Evidence	Defensive implication
F-01	Dashed-IP labels are being used as entity surrogates.	Medium	30 sampled labels match an IPv4 pattern with dashes instead of dots.	Taxonomy search and reporting can become noisy, inconsistent, and hard to govern.
F-02	Equivalent URL indicators and URL observables exist for all sampled values.	Medium	Live validation found at least one `Url` observable and one indicator for every normalized dotted IP.	Cleanup can proceed cautiously without losing the URL indicator layer.
F-03	Standalone `IPv4-Addr` modeling is not proven by the sampled validation.	Low	The validation returned URL observables for all values; it did not establish that first-class IP observables already exist.	If infrastructure-level pivoting is needed, create/link `IPv4-Addr` observables before removing labels.
F-04	The labels appear tied to ScreenConnect-style URL indicators in at least the leading sample.	Medium	Example: `https://209.99.185.184/Bin/ScreenConnect.ClientSetup.exe` with labels `connectwise`, `exe`, and `ua-wget`.	Hunt on URL and network telemetry, not on label strings.
F-05	The candidate is properly queued as manual review.	Low	Candidate `safe_mode: true`; `apply_mode: manual-review-only`.	Avoid broad automated label deletion until object relationships are exported and reviewed.

Normalized candidate coverage:

Dashed label	Normalized IP	URL observables found	Indicators found	Notes
`209-99-185-184`	`209.99.185.184`	1	1	URL/indicator present
`176-97-210-86`	`176.97.210.86`	1	1	URL/indicator present
`198-98-50-94`	`198.98.50.94`	3	3	URL/indicator present
`172-86-72-167`	`172.86.72.167`	2	2	URL/indicator present
`176-65-139-224`	`176.65.139.224`	1	1	URL/indicator present
`154-12-31-97`	`154.12.31.97`	1	1	URL/indicator present
`176-65-139-99`	`176.65.139.99`	20	20	Multiple URL/indicator objects
`107-182-128-220`	`107.182.128.220`	2	2	URL/indicator present
`176-65-139-119`	`176.65.139.119`	6	6	Multiple URL/indicator objects

`124-198-132-139`	`124.198.132.139`	2	2	URL/indicator present
`107-182-128-74`	`107.182.128.74`	2	2	URL/indicator present
`45-11-56-76`	`45.11.56.76`	2	2	URL/indicator present
`178-16-54-243`	`178.16.54.243`	1	1	URL/indicator present
`195-96-132-13`	`195.96.132.13`	11	11	Multiple URL/indicator objects
`45-156-87-91`	`45.156.87.91`	2	2	URL/indicator present
`156-238-236-16`	`156.238.236.16`	1	1	URL/indicator present
`209-99-190-86`	`209.99.190.86`	1	1	URL/indicator present
`160-30-18-123`	`160.30.18.123`	14	14	Multiple URL/indicator objects
`104-194-132-138`	`104.194.132.138`	2	2	URL/indicator present
`45-156-87-136`	`45.156.87.136`	1	1	URL/indicator present
`103-97-178-242`	`103.97.178.242`	1	1	URL/indicator present
`176-65-149-142`	`176.65.149.142`	12	12	Multiple URL/indicator objects
`198-98-55-166`	`198.98.55.166`	1	1	URL/indicator present
`209-99-188-149`	`209.99.188.149`	2	2	URL/indicator present
`80-94-92-170`	`80.94.92.170`	6	6	Multiple URL/indicator objects
`134-199-190-221`	`134.199.190.221`	3	3	URL/indicator present
`62-60-226-185`	`62.60.226.185`	2	2	URL/indicator present
`142-93-255-1`	`142.93.255.1`	3	3	URL/indicator present
`45-94-31-83`	`45.94.31.83`	1	1	URL/indicator present
`45-198-224-8`	`45.198.224.8`	5	5	Multiple URL/indicator objects

5. Threat Context

This candidate is primarily a threat-intelligence platform hygiene issue, but the values being normalized are operationally meaningful. The observed example URL pattern, `https://<ip>/Bin/ScreenConnect.ClientSetup.exe`, is consistent with infrastructure hosting a ConnectWise ScreenConnect installer. ScreenConnect is a legitimate remote monitoring and management product that is commonly abused when adversaries stage remote access tooling or social-engineer users into installing remote-control clients. This report does not attribute the candidate IPs to a specific actor, and the presence of a ScreenConnect URL alone is not proof of malicious activity. It does justify treating the URL and IP values as indicators requiring clean STIX representation.

The immediate risk is not that labels create new compromise exposure; it is that poor taxonomy modeling degrades the source of truth. Labels should express stable categories such as malware family, tool family, campaign name, source, handling tags, or analytic themes. Atomic observables such as IP addresses, URLs, ASNs, and CVEs should be represented as STIX domain objects or cyber observables so analysts can pivot, enrich, expire, score, and relate them consistently.

Modeling issue	Operational risk	Correct OpenCTI representation
IP stored as label string	Weak pivoting, inconsistent search, no indicator lifecycle	`IPv4-Addr` observable and/or indicator pattern
URL stored only as label	Loss of URL path, scheme, and context	`Url` observable and `[url:value = ...]` indicator
ASN stored as label	No relationship to provider/hosting context	Autonomous-system observable or identity/infrastructure relationship where supported
CVE stored as label	Duplicate vulnerability taxonomy	`Vulnerability` object with external references
Taxonomy used for atomics	Inflated label count and noisy reports	Curated labels reserved for analytic taxonomy

MITRE ATT&CK mapping for downstream defensive use is conditional because the task is not actor attribution. If telemetry shows users downloading or executing the referenced ScreenConnect client URLs, relevant techniques include:

Technique	Name	Why it may apply
T1105	Ingress Tool Transfer	Remote access installer retrieval from IP-hosted URL.
T1219	Remote Access Software	ScreenConnect/ConnectWise client installation or execution.
T1071.001	Application Layer Protocol: Web Protocols	HTTPS retrieval from direct-IP URLs.
T1583.006	Acquire Infrastructure: Web Services	Infrastructure staging assessment, if tied to adversary-owned hosting.
T1595	Active Scanning	Useful only if telemetry indicates scanning from or toward the IPs; not inferred from labels alone.

6. Detection and Hunting

Detection should use normalized IPs, URLs, and executable/download behavior. Do not build detections around OpenCTI label values such as `209-99-185-184`; those are curation artifacts, not network telemetry artifacts.

Coverage area	Data source	Analytic goal	Priority
Direct-IP URL download	Proxy, EDR network, DNS-less HTTP/S logs	Identify users or hosts retrieving tools from IP-literal URLs.	High
ScreenConnect installer execution	EDR process/file telemetry	Detect unexpected remote-access client installation.	High
Repeated connections to	Firewall, NetFlow, EDR	Find beaconing or remote-	Medium

candidate IPs	network	control sessions after installer retrieval.	
OpenCTI taxonomy drift	OpenCTI API export	Prevent reintroduction of atomic-observable labels.	Medium
Report/indicator relationship integrity	OpenCTI graph export	Confirm cleanup did not orphan indicators or source context.	Medium

Microsoft Defender XDR / KQL hunt for network activity to the normalized candidate IPs:

```
let CandidateIPs = dynamic([
    "209.99.185.184", "176.97.210.86", "198.98.50.94", "172.86.72.167", "176.65.139.224",
    "154.12.31.97", "176.65.139.99", "107.182.128.220", "176.65.139.119", "124.198.132.139",
    "107.182.128.74", "45.11.56.76", "178.16.54.243", "195.96.132.13", "45.156.87.91",
    "156.238.236.16", "209.99.190.86", "160.30.18.123", "104.194.132.138", "45.156.87.136",
    "103.97.178.242", "176.65.149.142", "198.98.55.166", "209.99.188.149", "80.94.92.170",
    "134.199.190.221", "62.60.226.185", "142.93.255.1", "45.94.31.83", "45.198.224.8"
]);
DeviceNetworkEvents
| where RemoteIP in (CandidateIPs)
| project Timestamp, DeviceName, InitiatingProcessFileName, InitiatingProcessCommandLine, RemoteIP,
RemotePort, RemoteUrl, ActionType
| order by Timestamp desc
```

Microsoft Defender XDR / KQL hunt for direct-IP ScreenConnect installer retrieval or execution:

```
let CandidateIPs = dynamic([
    "209.99.185.184", "176.97.210.86", "198.98.50.94", "172.86.72.167", "176.65.139.224",
    "154.12.31.97", "176.65.139.99", "107.182.128.220", "176.65.139.119", "124.198.132.139",
    "107.182.128.74", "45.11.56.76", "178.16.54.243", "195.96.132.13", "45.156.87.91",
    "156.238.236.16", "209.99.190.86", "160.30.18.123", "104.194.132.138", "45.156.87.136",
    "103.97.178.242", "176.65.149.142", "198.98.55.166", "209.99.188.149", "80.94.92.170",
    "134.199.190.221", "62.60.226.185", "142.93.255.1", "45.94.31.83", "45.198.224.8"
]);
let DownloadEvents = DeviceNetworkEvents
| where RemoteIP in (CandidateIPs) or RemoteUrl has_any (CandidateIPs)
| where RemoteUrl has_any ("ScreenConnect.ClientSetup.exe", "/Bin/", "ScreenConnect")
| project Timestamp, DeviceId, DeviceName, RemoteIP, RemoteUrl, InitiatingProcessFileName,
InitiatingProcessCommandLine;
let ExecutionEvents = DeviceProcessEvents
| where FileName has_any ("ScreenConnect.ClientSetup.exe", "ScreenConnect.ClientService.exe",
"ScreenConnect.WindowsClient.exe")
or ProcessCommandLine has_any ("ScreenConnect", "ConnectWise")
| project ExecTimestamp=Timestamp, DeviceId, DeviceName, FileName, ProcessCommandLine,
InitiatingProcessFileName, InitiatingProcessCommandLine;
DownloadEvents
| join kind=leftouter ExecutionEvents on DeviceId
| where isnull(ExecTimestamp) or ExecTimestamp between (Timestamp .. 2h)
| order by Timestamp desc
```

Splunk SPL hunt for proxy or firewall logs containing candidate IPs:

```
(index=proxy OR index=firewall OR index=edr) earliest=-30d
(dest_ip IN
("209.99.185.184", "176.97.210.86", "198.98.50.94", "172.86.72.167", "176.65.139.224", "154.12.31.97", "176.6
5.139.99", "107.182.128.220", "176.65.139.119", "124.198.132.139", "107.182.128.74", "45.11.56.76", "178.16.5
4.243", "195.96.132.13", "45.156.87.91", "156.238.236.16", "209.99.190.86", "160.30.18.123", "104.194.132.138
", "45.156.87.136", "103.97.178.242", "176.65.149.142", "198.98.55.166", "209.99.188.149", "80.94.92.170", "13
4.199.190.221", "62.60.226.185", "142.93.255.1", "45.94.31.83", "45.198.224.8")
OR url IN
("**209.99.185.184**", "**176.97.210.86**", "**198.98.50.94**", "**172.86.72.167**", "**176.65.139.224**", "**154.12.31
.97**", "**176.65.139.99**", "**107.182.128.220**", "**176.65.139.119**", "**124.198.132.139**", "**107.182.128.74**", "
**45.11.56.76**", "**178.16.54.243**", "**195.96.132.13**", "**45.156.87.91**", "**156.238.236.16**", "**209.99.190.86*
*", "**160.30.18.123**", "**104.194.132.138**", "**45.156.87.136**", "**103.97.178.242**", "**176.65.149.142**", "**198.9
8.55.166**", "**209.99.188.149**", "**80.94.92.170**", "**134.199.190.221**", "**62.60.226.185**", "**142.93.255.1**", "
```



```
*45.94.31.83*", "**45.198.224.8*"))
| stats count min(_time) as first_seen max(_time) as last_seen values(url) as urls values(user) as
users values(process_name) as processes by src_host dest_ip
| convert ctime(first_seen) ctime(last_seen)
| sort -count
```

OpenCTI hygiene query pattern for future review automation:

```
query LabelSample($first: Int!) {
  labels(first: $first) {
    edges {
      node {
        id
        value
        color
      }
    }
  }
  pageInfo { globalCount }
}
```

Use this query output to flag labels matching `^(?:\d{1,3}-){3}\d{1,3}\$`, normalize candidate values by replacing dashes with dots, and then search the normalized value across `stixCyberObservables` and `indicators` before cleanup.

7. Recommendations

Priority	Recommendation	Owner	Acceptance criteria
1	Approve the candidate as a valid label-normalization issue.	CTI platform owner	Candidate remains tagged `medium` and proceeds to controlled cleanup planning.
2	Export all relationships for the 30 label IDs before mutation.	OpenCTI operator	CSV/JSON export lists label ID, attached object IDs, object types, names, markings, creators, and reports.
3	Remove dashed-IP labels from indicators and URL observables after equivalence review.	OpenCTI operator	No candidate dashed-IP labels remain attached to reviewed objects; URL indicators and observables remain intact.
4	Create/link `IPv4-Addr` observables only where needed for infrastructure pivoting.	CTI analyst	Each created IP observable has source context, marking, confidence, and relationship to relevant URL/indicator/report.
5	Add a recurring taxonomy guardrail.	Automation owner	Hygiene job detects IP-like, ASN-like, and CVE-like labels and queues non-destructive candidates.
6	Preserve labels for analytic taxonomy only.	CTI team	House rule documented: labels represent themes, sources, malware/tool families, campaigns, or handling tags, not atomics.
7	Re-run the hygiene audit after cleanup.	OpenCTI operator	Follow-up audit shows reduced `ip_like_dash_labels`; no loss in indicator/observable counts

			outside expected label cleanup.
--	--	--	---------------------------------

Proposed safe cleanup sequence:

- 1. Export the 30 label objects and all attached entity relationships.
- 2. For each attached object, confirm the observable or indicator contains the dotted IP or URL value independently of the label.
- 3. If the only representation of an IP is a URL object and the IP is analytically important, create an `IPv4-Addr` observable and link it to the URL/indicator/report before label removal.
- 4. Remove the dashed-IP label relationship from reviewed objects; do not delete the underlying URL or indicator.
- 5. Delete the label object only after it has no remaining relationships and no active saved searches, dashboards, or workflow rules depend on it.
- 6. Re-run `opencti\_hygiene\_audit.py` and archive both before/after artifacts.

Indicators of Compromise

No source-validated malicious atomic indicators were disclosed for this work item. Lost Boys Cyber therefore treats this section as an IOC disposition record so downstream publication, OpenCTI hygiene, and detection-engineering workflows do not create placeholder blocklist entries.

Indicator Type	Value	Disposition
OpenCTI object / label IDs	See scoped object IDs and label table above	Normalize as CTI data-model context; do not treat taxonomy labels as network/blocklist IOCs.
Workflow context	threat-intel / opencti-curation	Use the report's detection and hunting logic for monitoring; do not promote non-malicious metadata into IOC-only controls.

8. References

- [1] Lost Boys Cyber OpenCTI hygiene audit artifact: `/home/lost0x01/services/opencti-deploy/audit/latest-opencti-hygiene-audit.json`.
- [2] Lost Boys Cyber OpenCTI cleanup candidate artifact: `/home/lost0x01/services/opencti-deploy/audit/latest-opencti-cleanup-candidates.json`.
- [3] Live validation evidence generated for this task: `/home/lost0x01/.hermes/kanban/workspaces/t\_2fe470e3/opencti\_ip\_label\_review\_evidence.json`.
- [4] OpenCTI local API endpoint used for validation: `http://127.0.0.1:8080/graphql`.
- [5] MITRE ATT&CK: T1105 Ingress Tool Transfer; T1219 Remote Access Software; T1071.001 Web Protocols; T1583.006 Web Services; T1595 Active Scanning.